BRADY S. HERWIG

Lynchburg, VA · 954-494-1314 · bradyherwig@outlook.com · bradyherwig.github.io

EDUCATION

Liberty University, Lynchburg, VA August 2024 – Expected May 2028

B.S. Computer Science, Data Science concentration · Accounting minor · GPA 3.64

Coursework: Advanced Machine Learning, Python and R for Data Science, Probability and Statistics, Discrete Mathematics

EXPERIENCE

Hyper Company LLC May 2026 – August 2026

AI Workflow Automation Intern

- Completed 25+ logic changes on production AI agents used for document workflows at insurance carriers and underwriting firms, raising extraction accuracy from 70% to over 98%.
- Led the UMR/Participation Schedule agent: automated extraction of UMR numbers, policy details, and syndicate allocations, cutting manual review on participation schedules from 100% to under 10%.
- Contributed to launching 3 agents covering document extraction, policy processing, and participation-schedule handling.
- Drove a July 2026 Florida surplus-lines (FSLSO) compliance update; traced syndicate-data issues across 5+ insurers and refined extraction logic before the effective date.

Outback Steakhouse, Davie, FL Oct 2023 – Aug 2024; May – Aug 2025

Server

- Managed up to 10 tables in a high-volume restaurant, coordinating kitchen, bar, and floor teams and resolving guest issues in real time.

PROJECTS

Consumer Complaint Topic Modeling KNIME · LDA · STM

CSCN 421, Advanced Machine Learning

- Fit LDA on 30,000 consumer-complaint narratives; compared 10-, 15-, and 20-topic models against a 15-topic STM and selected the 15-topic LDA for clearer, separable defect themes.
- Joined complaints to census region and found fuel-pump issues at 11.3% of Midwest complaints vs. 5.8% in the West, and power-steering issues at 10.5% vs. 5.8%.

Customer Segmentation R · K-Means · Hierarchical clustering · PCA

CSCN 321, Python and R for Data Science

- Engineered and standardized retail features, compared K-Means and hierarchical clustering with PCA, and chose the cluster count from silhouette scores and elbow plots.
- Wrote a Quarto report that turned the segments into recommendations a non-technical stakeholder could use.

Customer Churn Classification NumPy · KNIME · pure Python

Neural Network Fundamentals

- Built and trained a pure-NumPy single-hidden-layer network (forward/backprop, Glorot, full-batch GD, BCE) for Telco churn; KNIME preprocessing with stratified 70/30 split, one-hot encoding, and train-only normalization.
- Explored hyperparameters and diagnosed majority-class collapse; selected configuration achieving ~80% train / 80.8% test accuracy and analyzed class imbalance impact on minority-class recall.

TECHNICAL SKILLS

Languages: Python (pandas, scikit-learn, numpy), R, SQL (T-SQL), C++,

Analysis: Clustering, PCA, regression, time-series decomposition and forecasting, topic modeling (LDA), feature engineering, model evaluation

Tools: Jupyter, KNIME, PyTorch, Quarto, Git, Linux, VS Code

Applied: AI agent maintenance, document extraction, workflow automation, insurance filing compliance, Neural Networks

LEADERSHIP

Calvary Christian Academy, Fort Lauderdale, FL Sep 2023 – Apr 2024

Engineering Team Communications Lead

- Led communications for a 4-person engineering team and presented the final product to a board of engineers, earning the highest average project rating in school history.